```
      "* DL Register Area - 120Word
      "*
      "* DW00000 - DW00009 (PLS Use)
      "* DW00010 - DW00039 (Bit Use)
      "* DW00040 - DW00089 (Word Use)
      "* DW00090 - DW00119 (Timer)
      "*********************************************

      "# Program Name  :  1Scribe
      "# Functior      :
      "# Call Program  :
      "# Sub Program   :
      "# Func Program  :

      VAR;
          " Scribe Chuck
00000     LONG IDX_SCHY1 = 48;
00001     LONG IDX_SCHY2 = 49;
          " Scribe Z
00002     LONG IDX_CZ = 34;
          " Scribe Cutter
00003     LONG IDX_CX = 32;
00004     LONG IDX_CAMX = 33;
00005     LONG Axis_POS_CAMX = 0;
          " Scribe Shif
00006     LONG IDX_CS = 35;
00007     LONG Addr = 0;
          " Scribe Speed
00008     LONG Scribe_Speed_1st = C
00009     LONG Scribe_Speed_2nd = C
00010     LONG Scribe_Speed_3rd = C
00011     LONG SC_PRT_Addr_L = 0;
      END_VAR;

00012     Addr = IDX_CS * 50;

00013     SC_PRT_Addr_L = (ML15414 - 1) * 100;

          " Manual/Auto 구분
00014     DB00020 = 0;                                      " Manual Data
00015     DB00021 = 0;                                      " Scribe Complete Position 1 Cycle Selec
      t
00016     DB00022 = 0;                                      " Idling & Don't Cutter
00017     IF MB00004 == 1                                   " AutoRunning
00018         DB00020 = 1;                                  " Recipe Data
00019     ELSE;
00020         IF MB225580 == 1                              " Recipe Data
00021             DB00020 = 1;                              " Recipe Data
00022         IEND;
00023         IF MW00051 == 52;
00024             DB00021 = 1;                              " Scribe Complete Position 1 Cycle Selec
      t
00025         IEND;
00026     IEND;
00027     IF MB000800 | !MB000802 == 1;
00028         DB00022 = 1;                                  " Idling & Don't Cutter
00029     IEND;

          " Scribe Executing Flag
00030     IF !MB000800 & MB000802 == 1;
00031         MB15420F = 1;
00032     IEND;

          " Cutter Select
          " UPCZ - LOCZ
00033     IF !MB11420 & !MB11440 == 1;
00034         MB158800 = 1;                                 " Upper Cutter Z
00035         MB158801 = 0;                                 " Upper Roller Z
00036         MB158802 = 1;                                 " Lower Cutter Z
00037         MB158803 = 0;                                 " Lower Roller Z
00038     IEND;
          " UPCZ - LORZ
00039     IF !MB11420 & MB11440 == 1;
00040         MB158800 = 1;                                 " Upper Cutter Z
00041         MB158801 = 0;                                 " Upper Roller Z
00042         MB158802 = 0;                                 " Lower Cutter Z
00043         MB158803 = 1;                                 " Lower Roller Z
00044     IEND;
          " UPRZ - LOCZ
00045     IF MB11420 & !MB11440 == 1;
00046         MB158800 = 0;                                 " Upper Cutter Z
00047         MB158801 = 1;                                 " Upper Roller Z
00048         MB158802 = 1;                                 " Lower Cutter Z
00049         MB158803 = 0;                                 " Lower Roller Z
00050     IEND;

00051     DB000000 = 0;                                     " 1st Cutting Cmt
```

```
00052       DB000001 = 0;                                        " 2nd Cutting Cmt
00053       DB000002 = 0;                                        " 3rd Cutting Cmt
00054       DB000010 = 0;                                        " 1st Set Flag (0:Use 1:NotUse)
00055       DB000011 = 0;                                        " 3rd Set Flag (0:Use 1:NotUse)

00056       PLD [CX][CS][CZ][CAMX];

00057       IF DB00020 == 0                                      " Manual Data
00058           DB000010 = 1;                                    " 1st Set Flag (0:Use 1:NotUse)
00059           DB000011 = 1;                                    " 3rd Set Flag (0:Use 1:NotUse)
00060       ELSE;
00061           IF ML15292 == 0;                                 " 1st Cutter X Speed(UPCX)
00062               IF ML15294 == 0                              " 1st Cutter X Speed(LOCX)
00063                   DB000010 = 1;                            " 1st Set Flag (0:Use 1:NotUse)
00064               IEND;
00065           IEND;
00066           IF ML15296 == 0;                                 " 3rd Cutter X Speed(UPCX)
00067               IF ML15298 == 0                              " 3rd Cutter X Speed(LOCX)
00068                   DB000011 = 1;                            " 3rd Set Flag (0:Use 1:NotUse)
00069               IEND;
00070           IEND;
00071           IF DB00021 == 1                                  " Scribe Complete Position 1 Cycle Selec
      t
00072               DB000010 = 1;                                " 1st Set Flag (0:Use 1:NotUse)
00073               DB000011 = 1;                                " 3rd Set Flag (0:Use 1:NotUse
00074           IEND;
00075       IEND;

00076       DL00100 = ML15210 - ML15202                          " UPCX Cutting Distanc
00077       DL00102 = ML15212 - ML15204                          " LOCX Cutting Distance
00078       DL00104 = ML15222 - ML15218                          " UPCS Cutting Distance
00079       DL00106 = ML15224 - ML15220                          " LOCS Cutting Distance
00080       IF DL00100 < 0;
00081           DL00100 = DL00100 * -1;
00082       IEND;
00083       IF DL00102 < 0;
00084           DL00102 = DL00102 * -1;
00085       IEND;

            " 1st Cutting(Not) / 3rd Cutting(Not)
00086       IF DB000010 & DB000011 == 1
                " 2nd
                " Upper Scribe End Position Cal
00087           DL00310 = ML15210;
                " Lower Scribe End Position Cal
00088           DL00312 = ML15212;
                " Shift Y End Position Calc
00089           DL00314 = ML40030[Addr] + ML1522

                " Distance Calc
00090           DL00200 = DL00310 - ML15202                      " UPCX
00091           DL00202 = DL00312 - ML15204                      " LOCX
00092           DL00204 = ML15222 - ML15218                      " UPCS
00093           DL00206 = ML15224 - ML15220                      " LOCS

                " 1st
00094           DL00300 = DL00310;
00095           DL00302 = DL00312;
                " 3rd
00096           DL00320 = DL00310;
00097           DL00322 = DL00312;
00098       ELSE;
                " 1st Cutting(Not) / 3rd Cutting(Use)
00099           IF DB000010 == 1;
                    " 2nd
                    " Upper Scribe End Position Cal
00100               DL00310 = ML15210 - ML15288;
                    " Lower Scribe End Position Cal
00101               DL00312 = ML15212 - ML15290;
                    " Shift Y End Position Calc
00102               DF00424 = (DL00100 - ML15288) / DL00100 * DL00104;
00103               DF00426 = (DL00102 - ML15290) / DL00102 * DL00106;
00104               DL00434 = DF00424;
00105               DL00436 = DF00426;

00106               DL00314 = ML40030[Addr] + ML15218 + DL0043

                    " 3rd
                    " Upper Scribe End Position Cal
00107               DL00320 = ML15210;
                    " Lower Scribe End Position Cal
00108               DL00322 = ML15212;
                    " Shift Y End Position Calc
00109               DL00324 = ML40030[Addr] + ML15222;

                    " Distance Calc
00110               DL00200 = DL00310 - ML15202                  " UPCX
```

```
            DL00202 = DL00312 - ML15204                    " LOCX
            DL00204 = DL00414                              " UPCS
            DL00206 = DL00416                              " LOCS

            " 1st
            DL00300 = DL00310;
            DL00302 = DL00312;
        ELSE;
            " 1st Cutting(Use) / 3rd Cutting(Not)
            IF DB000011 == 1
                " 1st
                " Upper Scribe End Position Cal
                DL00300 = ML15202 + ML15284;
                " Lower Scribe End Position Cal
                DL00302 = ML15204 + ML15286;
                " Shift Y End Position Calc
                DF00404 = ML15284 / DL00100 * DL00104;
                DF00406 = ML15286 / DL00102 * DL0010
                DL00414 = DF00404;
                DL00416 = DF00406;

                DL00304 = ML40030[Addr] + ML15218 + DL00414;

                " 2nd
                " Upper Scribe End Position Cal
                DL00310 = ML15210;
                " Lower Scribe End Position Cal
                DL00312 = ML15212;
                " Shift Y End Position Calc
                DL00314 = ML40030[Addr] + ML1522

                " Distance Calc
                DL00200 = DL00310 - DL00300               " UPCX
                DL00202 = DL00312 - DL00302               " LOCX
                DL00204 = DL00314 - DL00304               " UPCS
                DL00206 = DL00316 - DL00306               " LOCS

                " 3rd
                DL00320 = DL00310;
                DL00322 = DL00312;
            ELSE;
                " 1st Cutting(Use) / 3rd Cutting(Use)
                " 1st
                " Upper Scribe End Position Cal
                DL00300 = ML15202 + ML15284;
                " Lower Scribe End Position Cal
                DL00302 = ML15204 + ML15286;
                " Shift Y End Position Calc
                DF00404 = ML15284 / DL00100 * DL00104;
                DF00406 = ML15286 / DL00102 * DL0010
                DL00414 = DF00404;
                DL00416 = DF00406

                DL00304 = ML40030[Addr] + ML15218 + DL0041

                " 2nd
                " Upper Scribe End Position Cal
                DL00310 = ML15210 - ML15288
                " Lower Scribe End Position Cal
                DL00312 = ML15212 - ML15290;
                " Shift Y End Position Calc
                DF00424 = (DL00100 - ML15288) / DL00100 * DL00104;
                DF00426 = (DL00102 - ML15290) / DL00102 * DL001
                DL00434 = DF00424;
                DL00436 = DF00426

                DL00314 = ML40030[Addr] + ML15218 + DL0043

                " 3rd
                " Upper Scribe End Position Cal
                DL00320 = ML15210;
                " Lower Scribe End Position Cal
                DL00322 = ML15212;
                " Shift Y End Position Calc
                DL00324 = ML40030[Addr] + ML1522

                " Distance Calc
                DL00200 = DL00310 - DL00300               " UPCX
                DL00202 = DL00312 - DL00302               " LOCX
                DL00204 = DL00314 - DL00304               " UPCS
                DL00206 = DL00316 - DL00306               " LOCS
            IEND;
        IEND;
    IEND;

    " Upper Cutter X
```

```
IF ML01142 == 0                                         " Upper Roller Use(0:Cutter 1:Roller)
    DL00500 = DL00300
    DL00510 = DL00310;
    DL00520 = DL00320;
ELSE;
    IF CW00001 == 1                                     " Scribe Direction (0: Forward / 1: Reve
rse)
        DL00500 = DL00300 + ML23746
        DL00510 = DL00310 + ML23746;
        DL00520 = DL00320 + ML23746;
    ELSE;
        DL00500 = DL00300 - ML23746
        DL00510 = DL00310 - ML23746;
        DL00520 = DL00320 - ML23746
    IEND;
IEND;
" Lower Cutter X
IF ML01144 == 0                                         " Lower Roller Use(0:Cutter 1:Roller)
    DL00502 = DL00302
    DL00512 = DL00312;
    DL00522 = DL00322;
ELSE;
    IF CW00001 == 1                                     " Scribe Direction (0: Forward / 1: Reve
rse)
        DL00502 = DL00302 + ML23748
        DL00512 = DL00312 + ML23748;
        DL00522 = DL00322 + ML23748;
    ELSE;
        DL00502 = DL00302 - ML23748
        DL00512 = DL00312 - ML23748;
        DL00522 = DL00322 - ML23748
    IEND;
IEND;

" 스크라이브 재시작 동작 중
DB000030 = 1;                                           " 1st Cutting Enabl

DB000031 = 1;                                           " 2nd Cutting Enabl
DB000032 = 1;                                           " 3rd Cutting Enable
IF MB047350 == 1                                        " Scribe Emergency Stop
    IF CW00001 == 1                                     " Scribe Direction (0: Forward / 1: Reve
rse)
        IF DL00500 > ML04910;
            DB000030 = 0;                               " 1st Cutting Enable
        IEND;
        IF DL00520 > ML04910;
            DB000031 = 0;                               " 2nd Cutting Enable
        IEND;
    ELSE;
        IF DL00500 < ML04910;
            DB000030 = 0;                               " 1st Cutting Enable
        IEND;
        IF DL00520 < ML04910;
            DB000031 = 0;                               " 2nd Cutting Enable
        IEND;
    IEND;
IEND;

" Speed Calc and Settin
DF00210 = DL00200 * DL00200                             " dx1
DF00212 = DL00202 * DL00202                             " dx2
DF00214 = DL00204 * DL00204                             " dy1
DF00216 = DL00206 * DL00206                             " dy2

" dx1^2 + dx2^2 + dy1^2 + dy2^2
DF00220 = DF00210 + DF00212 + DF00214 DF00216;
DF00222 = SQT(DF00220);
DF00224 = DF00210;
IF DF00224 < DF00212;
    DF00224 = DF00212;
IEND;
IF DF00224 < DF00214;
    DF00224 = DF00214;
IEND;
IF DF00224 < DF00216;
    DF00224 = DF00216;
IEND;
DF00226 = SQT(DF00224);

IF DF00222 == 0;
    DF00222 = 1;
IEND;
IF DF00226 == 0;
    DF00226 = 1;
IEND;
DF00228 = ML01124 * DF00222 / DF00226 * 60;
DL00230 = DF00228;
```

```
00234       DL00012 = ML10800[IDX_CX];
00235       DF00020 = DL00012 / 60 / ML01146 * 1000;
00236       DF00022 = DL00012 / 60 / ML01148 * 1000;
00237       DL00014 = DF00020;
00238       DL00016 = DF00022;

00239       IF DL00014 > 32767;
00240           DL00014 = 32767;
00241       IEND;
00242       IF DL00016 > 32767;
00243           DL00016 = 32767;
00244       IEND;

            " 1st Cutter X Speed(UPCX) < 1구간 Cutter X Speed(LOCX)
00245       IF ML15292 < ML15294
00246           Scribe_Speed_1st = ML15294 * 1.414 * 60;
00247       ELSE;
00248           Scribe_Speed_1st = ML15292 * 1.414 * 6
00249       IEND;

00250       Scribe_Speed_2nd = DL00230;

            " 3rd Cutter X Speed(UPCX) < 3rd Cutter X Speed(LOCX)
00251       IF ML15296 < ML15298;
00252           Scribe_Speed_3rd = ML15298 * 1.414 * 60;
00253       ELSE;
00254           Scribe_Speed_3rd = ML15296 * 1.414 * 6
00255       IEND;

00256       DL00030 = Scribe_Speed_1st;
00257       DL00032 = Scribe_Speed_2nd;
00258       DL00034 = Scribe_Speed_3rd;

00259       PFORK Label1 Label2 Label3
            Label1:

00260           IF (DB00020 & !DB00021 & !DB00022) ==             " Recipe Data
00261               ML04850 = ML43412 * 2 * 60;

00262               IOW DB000000 == 1                             " 1st Cutting Cmt

                    " 2nd
00263               MW15520 = 7;                                  " Torque Setting
00264               MW15525 = 9;                                  " Axis Select
00265               MSEE MPS253;                                  " CZ Torque Set

00266               ML16008 = ML15226                             " [UPCZ] Upper Cutter Z Moving Position
        Write

00267               IOW MB045778 == 1                             " Cutter Z Move Permission

00268               MVS [CZ]ML16008 FML04850;

00269               IF DB000011 == 0                              " 3rd Set Flag (0:Use 1:NotUse)
00270                   IOW DB000001 == 1                         " 2nd Cutting Cmt

                        " 3rd
00271                   MW15520 = 11;                             " Torque Setting
00272                   MW15525 = 9;                              " Axis Select
00273                   MSEE MPS253;                              " CZ Torque Set

                        " 3rd Cutter Z Scribe Infeed Volume(um)
                        " Upper Head

                        " UPCZ, UPRZ
00274                   ML15274 = ILA116 + GL100062[SC_PRT_Addr_L];

00275                   ML16008 = ML15274                         " [CZ] Upper Cutter Z Moving Position Wr
        ite

00276                   IOW MB045778 == 1                         " Cutter Z Move Permission

00277                   MVS [CZ]ML16008 FML04850;
00278               IEND;

00279               IOW DB000002 == 1                             " 3rd Cutting Cm

00280               MSEE MPS255;                                  " CZ Torque Reset
00281           IEND;

00282           JOINTO LabelX;

            Label2:

00283           FMX tDL00012;
00284           IAC tDL00014;
```

```
        IDC tDL00016;

        ACCMODE M4;

        S{MB047351} = DB00020                           " Scribe Run Check

        " 1st Cutting(Not) / 3rd Cutting(Not)
        IF DB000010 & DB000011 == 1;
            DB000000 = 1;

            IF DB000031 == 1                            " 2nd Cutting Enable
                MVS [CX]DL00510 [CS]DL00314 FDL00032 FE0;
            IEND;

            IOW IBA00C1 == 1;
            DB000001 = 1;
            DB000002 = 1;
        ELSE;
            " 1st Cutting(Not) / 3rd Cutting(Use)
            IF DB000010 == 1;
                DB000000 = 1;

                IF DB000031 == 1                        " 2nd Cutting Enable
                    MVS [CX]DL00510 [CS]DL00314 FDL00032 FEDL00034;
                IEND;

                DB000001 = 1;

                IF DB000032 == 1                        " 3rd Cutting Enable
                    MVS [CX]DL00520 [CS]DL00324 FDL00034 FE0;
                IEND;

                DB000002 = 1;
            ELSE;
                " 1st Cutting(Use) / 3rd Cutting(Not)
                IF DB000011 == 1

                    IF DB000030 == 1                    " 1st Cutting Enable
                        MVS [CX]DL00500 [CS]DL00304 FDL00030 FEDL00030;
                    IEND;

                    DB000000 = 1;

                    IF DB000031 == 1                    " 2nd Cutting Enable
                        MVS [CX]DL00510 [CS]DL00314 FDL00032 FE0;
                    IEND;

                    DB000001 = 1;
                    DB000002 = 1;
                ELSE;
                    " 1st Cutting(Use) / 3rd Cutting(Use)
                    IF DB000030 == 1                    " 1st Cutting Enable
                        MVS [CX]DL00500 [CS]DL00304 FDL00030 FEDL00030;
                    IEND;

                    DB000000 = 1;

                    IF DB000031 == 1                    " 2nd Cutting Enable
                        MVS [CX]DL00510 [CS]DL00314 FDL00032 FEDL00034;
                    IEND;

                    DB000001 = 1;

                    IF DB000032 == 1                    " 3rd Cutting Enable
                        MVS [CX]DL00520 [CS]DL00324 FDL00034 FE0;
                    IEND;

                    DB000002 = 1;
                IEND;
            IEND;
        IEND;

        MB047351 = 0;                                   " Scribe Run Chec
        MB047350 = 0;                                   " Scribe_StopRetr_


        ACCMODE M0;

        JOINTO LabelX;

    Label3:

        " UCAMX Escape Position
        ML16066 = ML43330                               " UCAMX Waiting Position
        IOW MB04577A == 1                               " UCAMX Moving Permission
        VEL [CAMX]ML43312;
        MOV [CAMX]ML16066;
```

```
          IOW MB046010 == 1                                      " UCAMX Waiting Position Chec

          JOINTO LabelX;
      LabelX: PJOINT;

      MB15420F = 0;                                              " Scribe Executing Flag

  RET;
```